

Project A

Diploma of IT Programming 2

SEMESTER 1 2026

Weight: 20%

DUE : Week 5 (written task), Week 7 (final submission)

ASSESSMENT TASK 1

TASK DESCRIPTION

This project requires students to create an object oriented design and coded solution for a chosen application, using all the concepts taught in the subject up to and including week 6.

You **MUST** complete this assessment in a group of 2 or 3 students belonging to the same tutorial. You will need to spend approx. 10 hours together working on the assessment (outside of tutorial hours). All group members must put in an equal amount of work and effort; individual penalties will apply for lesser-contributing members.

OBJECTIVES

1. Evaluate if a software solution is well-designed and fit for purpose.
2. Design a well-constructed object oriented solution from a specification.
3. Demonstrate a working knowledge of list and map data structures.
4. Use inheritance to improve the system design.

BRIEF

This is a group assessment to evaluate your understanding of Java programming concepts taught so far in IPRO002. It includes an in-class writing component. The final submission will be composed of three (3) parts: the scenario, the code and explanation (including proof of testing).

DEVISE SCENARIO AND BASIC USE-CASES

During the beginning of tutorial A in week 5 your group will be given a topic e.g. a café, or a library. Then sitting apart from your other group members you will be required to:

- Write a single paragraph of 4 to 5 sentences that describes a business scenario for the given topic.
- Describe 5 use-cases for a computer program that will serve needs of the business described in your scenario.

This task will be performed and graded **individually** under exam conditions with pen and paper. There will be no access to any electronic devices during this task.

After submission your tutor will retain copies of your work, and it will be returned to you in the following lab/tutorial B session of that week.

FINALISE SCENARIO AND USE-CASES

As a group, take stock of each member's submission and come to an agreement on the final form of the scenario and use-cases. Groups of 2 students are required to have a minimum of 4 use-cases while groups of 3 need a minimum of 6 use-cases.

Note: Your use-cases cannot be overly simple e.g. adding, removing and modifying items in an order **do not** count as 3 separate use-cases but one. **If you have any doubts about your chosen use-cases then consult your teacher as early as possible.**

THE DESIGN

Now identify the major functionality that your application must provide to meet the use-cases determined earlier. At a minimum, your program needs to respond to user-input from the keyboard to:

- add and remove objects to collections
- retrieve a specific object from a collection
- sort a collection in appropriate ways and display its objects
- create desired sub-collections from a larger collection e.g. extracting songs from a particular genre from an audio track library.

You are **required** to create a simplified class diagram for your application, which shows the attributes/methods for each class, inheritance relationships and interface implementations. Your application should aim to have:

- between 6 and 8 classes for a group of 2 members or

- between 9 and 10 classes for a group of 3 members

The classes should include: Main/driver class, parent class (at least 1), child classes (at least 2) and at least 1 enum. You will explain how your design will provide the needed functionality for the application.

THE CODE

You will write the code for the application using the design. Make sure you follow the guidelines set out in the subject e.g. use camelCase for objects, LetterCase for classes, all classes have toString(). Public mutators should be used sparingly etc.

At the top level, your solution must present a text-based menu to the user as shown in week 2. Your solution should also use **both** interface and implementation inheritance. Also look for opportunities to apply method overriding and overloading.

Be sure to adhere to standard Java naming conventions.

As with Programming 1 last semester, you must only use code constructs taught in weeks 1 to 6 inclusive. Using constructs outside of week 1 to 6 will substantially lower the marks you receive.

THE EXPLANATION

You are **required** to explain how you came to your chosen scenario and use-cases, referencing your initial written pieces from week 5.

You must also explain the submitted Java code and any problems you encountered and how you fixed them. At minimum you need to identify :

- Inheritance – including why inheritance was used in this/these class(es)
- Interfaces – including why particular classes implemented them
- Overloaded methods – why overloading was used in each case
- Overriding methods – why overriding was used in each case
- Enum classes – why enum was used in each case
- Polymorphism – how it affected the code

You also need to include screen shots of your code running with VS Code.

Please double check that your code runs as intended and your submission document contains the updated version of all code prior to submission.

LIST OF CONTRIBUTIONS

Your group is required to list the contributions of each student e.g.

Student A:

- Wrote classes X, Y and Z
- Wrote sections B and C in the explanation

Student B:

- Wrote classes D, E
- Wrote test cases

Etc.

COVER SHEET

Your group is required to fill out the cover sheet listed below and attached as the first page of your word document submission. **You must specify the % contribution of each student.**

ASSESSMENT CRITERIA

CRITERIA	WEIGHT	SLOs / LLOs	PLOs
Class relationships	15%	SLO002, SLO004	A1, E1
Functional specifications	25%	SLO001, SLO002, SLO003	A1, B1
Design rules	25%	1, 3	A1, E1
Write a scenario and explanation, identifying elements and justifying coding choices	5%	SLO004	A1, B1, C1
Write cohesively and coherently (in-class)	15%	LLO004	C1
Use content specific lexicogrammar (in-class)	15%	LLO004	C1

SLOs: subject learning outcomes

PLOs: program learning outcomes

LLOs: language learning outcomes

Please refer to the next page for the rubric definition for each criteria and grade.

**I PRO002 – PROGRAMMING A**
GROUP ASSIGNMENT**ASSIGNMENT COVER PAGE****Semester 3, 2025**

Student ID	Student name	% contribution to group effort

Declaration

I confirm that the material contained in this assignment is my own work and has not been submitted in any other subject or course. I am aware of plagiarism and its penalties, and have acknowledged all material and sources used in the preparation of this assignment.

Date:

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Signature:

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		HD	D	C	P	F3	F2	F1	
1	Class relationships	All classes, relationships and inheritance required to complete the project have been identified. Demonstrates a clear and comprehensive understanding of OO concepts and how they should be applied to produce a working design.	All major classes, relationships and inheritance have been identified. Demonstrates a clear understanding of OO concepts and how they should be applied to produce a working design	Most behaviour has been identified correctly and placed in correct classes. Some relationships missing or inheritance missing. Demonstrates moderate understanding of OO concepts and how they should be applied to produce a working design	Some classes and relationships have been identified and some behaviour has been placed in correct classes. Some correct inheritance. Demonstrates some understanding of OO concepts and how they should be applied to produce a working design	Some classes have been identified. Few relationships. Minimal or incorrect inheritance. Demonstrates little understanding of OO concepts and how they should be applied to produce a working design	Most classes are missing or wrong. Little relationships/inheritance	Only one class has been identified. No correct inheritance or relationships.	No attempt
2	Functional specifications	The program works correctly in every way and meets all of the functional specifications. Lists and looping constructs are correctly and naturally used in both a defined class and in the scenario.	The program works and produces correct results except in one area. It also meets most of the other specifications. Lists and looping constructs are correctly and naturally used in either a defined class or in the scenario (but not both).	The program mostly works and produces mostly correct results. It also meets most of the other specifications. Lists and looping constructs are present.	The program is producing some incorrect results or a part of the program is incomplete. Lists and looping constructs are not present or if present, contains errors.	The program is producing mostly incorrect results or a most of the program is incomplete.	The program is producing all incorrect results or mostly incomplete.	The program does not run; Some code is valid	Program do

3	Design rules	The code is exceptionally well organised and variable names chosen clearly explain what the code is accomplishing and how. Java naming conventions always used. All Java constructs are from week 1 to 6.	The code is easy to follow and well organised and variable names chosen are very useful in understanding what the code does and how it is accomplished. Java naming conventions mostly used. Most Java constructs are from week 1 to 6.	The code is fairly easy to follow and variable names chosen are useful in understanding the code. Java naming conventions sometimes used. Java constructs are generally from week 1 to 6.	The code is a little hard to read and variable names chosen are somewhat useful in understanding the code. Java naming conventions used at least once. Some Java constructs are from week 1 to 6.	The code is hard to read and variable names chosen at least once help understand what they are used for. Java naming conventions not used. A few Java constructs are from week 1 to 6.	The code is poorly organised and very difficult to follow. Variable names chosen do not help understand what they are used for. Reader must guess what variables are for. Java naming conventions not used. Only 1 Java construct is from week 1 to 6.	The code is poorly organised and very difficult to follow. Variable names chosen use names that will not compile e.g. with spaces. Java naming conventions not used. No Java construct is from week 1 to 6.	No valid code
4	Write a scenario and explanation, identifying elements and justifying coding choices.	Scenario is engaging, unique and well thought out. Explanation correctly identifies all required elements and convincingly justifies coding choices. Work is comprehensive.	Scenario is unique and well thought out. Explanation correctly identifies almost all required elements and justifies most coding choices. Work is mostly comprehensive.	Scenario is a unique idea and mostly explained. Explanation correctly identifies most required elements and justifies some coding choices. Work is mostly comprehensive.	Scenario is a mostly unique with some parts explained. Explanation correctly identifies some required elements and justifies a few coding choices. Work is generally comprehensive.	Scenario is somewhat unique with some parts perhaps explained. Explanation correctly identifies a few required elements and justifies very few coding choices.	Scenario is flawed or inadequate. Explanation correctly identifies very few required elements or justifies very few coding choices. Work is substandard.	Scenario is confused or copied from another source. Explanation does not identify any elements or code choices, or is copied from another source. Work is not acceptable.	No submission

WT002	Write cohesively and coherently (in-class)	ALL aspects/features must be met. Requires minimal effort to follow the text because 1. paragraphs are reasonably well-structured and have reasonably clear and appropriate topic sentences, 2. information is usually logically connected, and 3. ideas are usually linked smoothly.	At least ONE, but not all aspects/features of HD are met; all are above P.	At least ONE aspect/feature is above P (but not necessarily HD standard); all others are of P standard.	ALL aspects/features must be met. Requires some effort to follow because, 1. while there are paragraphs, some lack topic sentences 2. some information is not logically connected, 3. and ideas are only generally linked smoothly.	Not all aspects of P grade were achieved; ... OR ... task, as outlined in the brief, was not addressed.	...	...	No submission
WT003	Use content specific lexicogrammar	ALL aspects/features must be met: • Uses a wide range of content specific vocabulary, mostly appropriately. • May make errors in lexicogrammar, but they rarely impede communication.	At least ONE but not all aspect/features of HD are met. ALL others are above P.	At least ONE aspect/feature is above P; ALL others are P standard.	ALL aspects/features must be met: • Uses a limited range of discipline-specific vocabulary, generally appropriately. • Makes errors in lexicogrammar which sometimes impede communication.	Not all aspects of P grade are achieved; ... OR ... task requirements, as stated in the assessment brief, were not met.	...	...	No submission